

DBMS WEEKLY TASK 2

Project Title: Identifying Data Requirements for AJIO Products Order Booking Database System

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About Primary Key

A Primary Key (PK) is a unique identifier assigned to every record in a database table. It helps the database distinguish one record from another and ensures that duplicate records are not created. A primary key cannot contain NULL values and is commonly used to establish relationships between multiple tables.

Example:

Every product is assigned a unique ProductID such as 1001. Even if two products have the same name, their ProductID will always be different.

Main Data Entities

- 1. Customer:** CustomerID (PK), FirstName, LastName, Email, Password, Phone, DateOfBirth, RegistrationDate, AccountStatus, LoyaltyPoints.
- 2. Address:** AddressID (PK), CustomerID, AddressType, StreetAddress, City, State, Pincode, IsDefault.
- 3. Category:** CategoryID (PK), CategoryName, Description, ParentCategoryID, IsActive.
- 4. Product:** ProductID (PK), ProductName, Description, CategoryID, BrandName, Color, Size, Price, StockQuantity, ReorderLevel, IsActive.
- 5. ProductImage:** ImageID (PK), ProductID, ImageURL, IsPrimary.
- 6. Cart:** CartID (PK), CustomerID, ProductID, Quantity, AddedDate.
- 7. Order:** OrderID (PK), CustomerID, OrderDate, OrderStatus, TotalAmount, PaymentMethod, ShippingAddressID.
- 8. OrderItem:** OrderItemID (PK), OrderID, ProductID, Quantity, UnitPrice, TotalPrice.
- 9. Payment:** PaymentID (PK), OrderID, Amount, PaymentDate, PaymentStatus, TransactionID.
- 10. Shipment:** ShipmentID (PK), OrderID, CourierName, TrackingNumber, ShipmentDate, DeliveryStatus.
- 11. Review:** ReviewID (PK), CustomerID, ProductID, Rating, ReviewContent, ReviewDate.
- 12. Wishlist:** WishlistID (PK), CustomerID, ProductID, AddedDate.
- 13. AuditLog:** LogID (PK), UserID, ActionType, TableName, OldValue, NewValue, ActionDate.

Additional Data Requirements

Foreign Key Links: Address→Customer, Product→Category, ProductImage→Product, Cart→Customer & Product, Order→Customer & Address, OrderItem→Order & Product, Payment→Order, Shipment→Order, Review→Customer & Product, Wishlist→Customer & Product, Category→Category (self-reference).

Order Status Values: Pending, Confirmed, Shipped, Delivered, Cancelled.

Payment Status Values: Pending, Successful, Failed, Refunded.

Delivery Status Values: Pending, In Transit, Delivered, Returned.

Payment Methods: Card, UPI, COD, Wallet.

Key Business Rules

One customer can have multiple addresses, multiple orders, and multiple reviews.

One product can have multiple images and multiple reviews, but belongs to one category.

One order can have multiple order items, but exactly one payment and one shipment record.

Stock quantity cannot be negative, and a low stock alert is triggered when stock falls below the reorder level.

Customer accounts are marked inactive rather than deleted, and all admin actions are logged in the AuditLog.

Conclusion

The AJIO products order booking database is organized using well-defined entities and primary keys to ensure accurate data management, efficient relationships, secure transactions, and reliable inventory, order, payment, and delivery tracking.